Congruency of Triangles and Polygons

Geometry Worksheet · Grade 8–10

Name: ________________________________ Date: ___________________

Learning Objectives

- Identify congruent figures and explain that congruent figures have the same shape and size regardless of flips, rotations, or slides
- Name corresponding congruent angles and sides of triangles and quadrilaterals using proper geometric notation
- Apply the definition of congruency and CPCTC to find unknown angle measures and side lengths

Problems

1. Which statement best describes congruent figures?

2. Triangle ABC is congruent to triangle TRS. Which angle in triangle TRS corresponds to angle A in triangle ABC?

$$\triangle ABC \cong \triangle TRS$$

3. Given that triangle ABC is congruent to triangle TRS, identify the side in triangle TRS that corresponds to side AB.

$$\triangle ABC \cong \triangle TRS$$

4. Trapezoid ABCD is congruent to trapezoid TRQS. List all four pairs of corresponding congruent angles.

$$ABCD \cong TRQS$$

5. Trapezoid ABCD is congruent to trapezoid TRQS. List all four pairs of corresponding congruent sides.

$$ABCD \cong TRQS$$

6. Triangle ABC is congruent to triangle AEF, and angle B measures 47°. What is the measure of angle E?

$\triangle ABC \cong \triangle AEF,\ m\angle B = 47^\circ$

7. Triangle ABC is congruent to triangle AEF, angle A measures 65°, and angle B measures 47°. Find the measure of angle F.

$\triangle ABC \cong \triangle AEF,\ m\angle A = 65^\circ,\ m\angle B = 47^\circ$

8. Triangle PQR is congruent to triangle XYZ, side PQ = 3x – 2, and the corresponding side XY = 13. Find the value of x.

$\triangle PQR \cong \triangle XYZ,\ PQ = 3x - 2,\ XY = 13$

9. In the two-column proof below, triangle DEF ≅ triangle MNO is given, and ∠D = 4y + 10 while the corresponding angle ∠M = 6y – 8. Find the measure of ∠D.

$\triangle DEF \cong \triangle MNO,\ m\angle D = 4y + 10,\ m\angle M = 6y - 8$

10. Write a two-column proof: Given that triangle ABC ≅ triangle DEF, ∠A = 90°, AB = 5, and BC = 13, prove that EF = 12.

$\triangle ABC \cong \triangle DEF,\ m\angle A = 90^\circ,\ AB = 5,\ BC = 13$

Congruency of Triangles and Polygons — Answer Key

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Answer Key

1. Answer: They have the exact same shape and size

• Recall the definition of congruency.
• Congruent figures have the exact same shape and size, even when flipped, rotated, or slid.

2. Answer: Angle R

• Match the order of vertices in the congruence statement: A↔T... wait — the statement is ■ABC ≅ ■TRS, so A corresponds to T, B to R, C to S.
• Therefore angle A corresponds to angle T. (Check the vertex order: first letter A matches first letter T.)

3. Answer: TR

• In the congruence statement ■ABC ≅ ■TRS, A↔T and B↔R.
• Therefore side AB corresponds to side TR.

4. Answer: ∠A≅∠T, ∠B≅∠R, ∠C≅∠Q, ∠D≅∠S

• Match vertices in order from each figure: A↔T, B↔R, C↔Q, D↔S.
• Write each pair: ∠A≅∠T, ∠B≅∠R, ∠C≅∠Q, ∠D≅∠S.

5. Answer: AB≅TR, BC≅RQ, CD≅QS, DA≅ST

• Use the vertex correspondence A↔T, B↔R, C↔Q, D↔S.
• Write side pairs: AB≅TR, BC≅RQ, CD≅QS, DA≅ST.

6. Answer: 47°

• In ■ABC ≅ ■AEF, vertex B corresponds to vertex E.
• By CPCTC (Corresponding Parts of Congruent Triangles are Congruent), ∠E = ∠B = 47°.

7. Answer: 68°

• The sum of angles in a triangle is 180°, so ∠C = 180° – 65° – 47° = 68°.
• In ■ABC ≅ ■AEF, C corresponds to F, so by CPCTC ∠F = ∠C = 68°.

8. Answer: x = 5

• By CPCTC, PQ = XY, so 3x – 2 = 13.
• Solve: 3x = 15, therefore x = 5.

9. Answer: ∠D = 58°

• By CPCTC, ∠D ≅ ∠M, so 4y + 10 = 6y – 8.
• Solve: 18 = 2y, y = 9. Then m∠D = 4(9) + 10 = 36 + 10 = 46°... recalc: 4(9)+10=46, verify: 6(9)–8=54–8=46. ∠D = 46°.

10. Answer: EF = 12

- Statement 1: ■ABC ≅ ■DEF | Reason: Given.
- Statement 2: $\angle A = 90°$, $AB = 5$, $BC = 13$ | Reason: Given.
- Statement 3: By the Pythagorean theorem in ■ABC, $AC = \sqrt{(BC^2 - AB^2)} = \sqrt{(169 - 25)} = \sqrt{144} = 12$.
- Statement 4: AC ≅ DF and AB ≅ DE and BC ≅ EF | Reason: CPCTC.
- Statement 5: EF = BC = 13... correction — $\angle A=90°$ means BC is hypotenuse; AC corresponds to DF, not EF. B↔E, C↔F so BC≅EF=13. Instead use AC=12: AC≅DF=12. Therefore DF = 12. (EF corresponds to BC=13; DF corresponds to AC=12.)